Drift to Default: Brightness Drift in Autoregressive Video Diffusion Is Regression to a Prompt-Set Level

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Abstract

Autoregressive (causal) video diffusion models generate long videos chunk by chunk and drift over tens of seconds: the picture darkens, saturates or loses detail. We study this drift on Self Forcing (Wan2.1-1.3B) and LongLive with the design tools of clinical research: paired seeds, held-out prompts, mixed-effects models, cluster-bootstrap intervals and preregistered analyses. Across about **6,400** generated videos (mostly 31 s) we find: (i) anchoring the first frames in the KV cache (an attention sink) slows colour drift strongly, and a small rolling window drifts fastest without it (preregistered, 2,000 videos); (ii) brightness drift is *two-way regression to a default level*: brighter openings darken and darker ones brighten, only ≈22% of the opening's deviation survives 31 s, and the default is set by the text prompt (sunny 43, overcast 34, dusk 19, night 16 L^*) and differs between models (33 vs 40 for the same prompt) while the regression slope does not (−0.69 vs −0.74); (iii) a training-free, content-aware correction that transfers the per-channel mean tone of an anchor-only prediction at one early denoising step reduces drift without harming structure (preregistered confirmation on 40 held-out prompts) and transfers to LongLive at half the effect; (iv) in a preregistered test, prompts written from an off-the-shelf VLM caption of the opening did *not* reduce brightness drift on either model, although they kept 0.10–0.15 more of the opening's content (DINOv2 similarity); the captioner's lighting words are poorly calibrated to actual brightness. (v) In a second preregistered test, choosing the lighting description from the opening's measured brightness, with levels taken from (ii), reduced brightness drift against the generic prompt on Self Forcing (−26%, Holm $p = 0.013$); combined with the correction it cut brightness drift by 44% and colour shift by 39%. On LongLive the effect was smaller and not significant, because the calibration was estimated on Self Forcing. We also report what did not hold, and two evaluation pitfalls: the GPU architecture changes bf16 outputs, breaking seed pairing, and a popular image-quality metric is insensitive to this drift.

1 Introduction

Causal video diffusion models such as Self Forcing [1] distil a bidirectional video diffusion model into a student that generates a few latent frames at a time, attending to its own past through a KV cache. This makes long, streaming generation possible and is the basis of interactive "world models". Over long horizons the outputs drift. Most follow-up work proposes a fix and shows a few qualitative examples or one aggregate score from a single sample per prompt, without uncertainty.

This report treats drift as a measurement problem. We ask three questions and answer each with a design whose analysis plan was written down before the data existed where it mattered:

1. How do the two standard context strategies, a rolling window and an attention sink, affect drift? (§3)

2. What is drift? We find that brightness drift is regression to a default level set by the prompt, on two models. (§4)

3. Can it be reduced without training? A content-aware correction works and transfers (§5); describing the opening with a free-form caption does not (§6), but a lighting description calibrated to measured brightness does (§7).

No model was trained. Everything is inference on public weights, run on a university cluster. Code, prompts, preregistrations and per-experiment summaries are at `https://github.com/ElaineYilinW/drift-to-default`.

2 Setup

Models. Self Forcing (DMD checkpoint, Wan2.1-T2V-1.3B, 4 denoising steps, chunks of 3 latent frames = 12 video frames, 16 fps) with a rolling KV window of w latent frames and s permanently kept "sink" frames; default w=21. LongLive 1.0 [2] (the same base with a rank-256 LoRA, official w=12, s=3). Videos have 501 frames (≈31 s).

Reference-based drift metrics. For every prompt, 10 fresh 5-second generations define the model's own "normal" output. **D** is the Mahalanobis distance of each frame's colour statistics (Lab means, saturation, log sharpness) from this reference; **K** is the DINOv2 nearest-neighbour distance of a frame to the reference frames, divided by the reference's leave-one-seed-out median (so $K \approx 1$ means "as typical as a fresh start"); z_L is the signed brightness z-score. Both D and K are averaged over the last 2 seconds. For experiments that start from a held image, drift is measured relative to that opening: ΔL_{end}, the change of mean L* between the opening (8 frames) and the last 2 seconds, and ΔE_{end}, the Lab distance.

Statistics. Seeds are paired across arms (same initial noise and random stream). Effects are estimated with linear mixed models (random intercepts for prompt and prompt×seed) or as paired differences averaged over seeds, with prompt- or opening-cluster bootstrap confidence intervals and Holm correction for the primary family. Three experiments were preregistered with SHA-256 hashes of the analysis code (§3, §5, §6); the rest is labelled exploratory.

A pitfall: the GPU changes the video. With bf16, the same seed gives bit-identical videos within a GPU architecture but different videos on A40, on A100/A800 and on L40S/RTX 6000 Ada. On a shared cluster, about 70% of the "paired" comparisons in our first experiment ran on different architectures. This does not bias the comparison, but it loses the variance reduction that pairing is meant to give. All later experiments run every arm of a prompt in one job on one GPU.

3 Context strategies (preregistered)

We crossed window $w \in \{12, 21\}$ × sink $s \in \{0, 1, 3\}$ × RoPE re-alignment of the sink ∈ {no, yes} (10 strategies) over 40 prompts (20 MovieGen, 20 driving) × 4 seeds.

- Anchoring 3 frames cuts colour drift: D, sink 3 vs 0, −7.56 [-8.32, -6.81]; K −0.90 [-1.04, -0.76]. There is a dose effect ($3 > 1 > 0$).

- The window matters only without a sink (interaction: D +5.71 [4.21, 7.22], K +0.67).

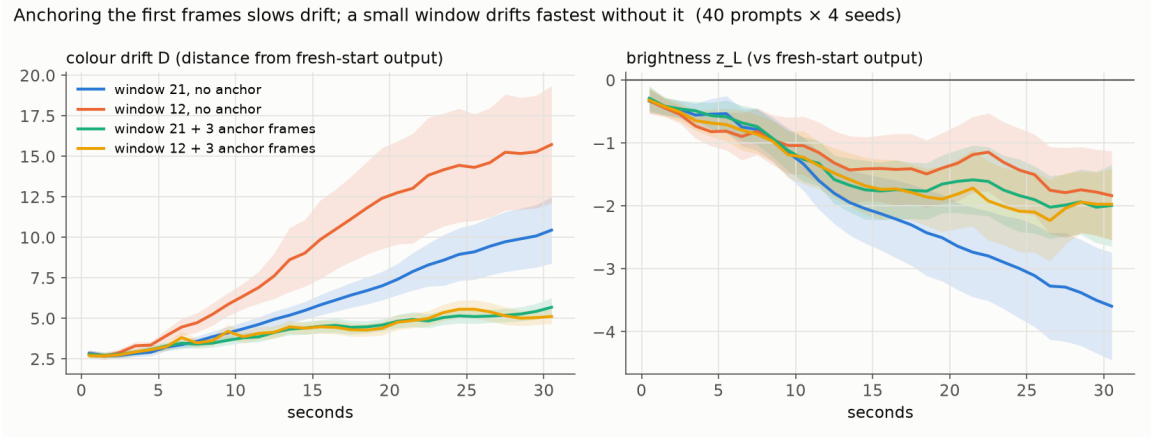


Figure 1: Colour drift D and DINOv2 drift K at the end of 31-second videos for 10 context strategies (Self Forcing, 40 prompts × 4 seeds; means with prompt-cluster bootstrap 95% CIs).

- Re-aligning the sink's positions, as Rolling/Deep Forcing do, did not help colour (D -0.03 [-0.79, 0.72]) and made K worse ($+0.36$ [0.22, 0.50]). Those methods were trained or designed for it; Self Forcing was not.
- Drift is slowed, not stopped: D rises roughly linearly to 31 s. Over-saturation disappears with a sink, but darkening does not ($z_L < 0$ in 78–88% of anchored videos).
- MUSIQ, the image-quality model behind VBench's imaging-quality score, is nearly flat across strategies (66–69). It does not see this drift.
- The seed variance of end-of-video drift was small next to the prompt variance (0.2 vs 13.7). More prompts buy more precision than more seeds.

4 Drift is regression to a prompt-set default

Held openings. To separate drift from what the prompt asks for, we held one image for 0.5 s as the first chunk and let the model continue it for 31 s with a generic driving prompt. Openings were 20 NAVSIM [7] front-camera frames and 20 first frames generated by the model itself. The change in brightness is a linear function of the opening's brightness (Fig. 2a):

$$\Delta L_{\text{end}} = 22.8 - 0.69\, L_{\text{open}}, \qquad R^2 = 0.82, \quad \text{slope } [-0.81, -0.57].$$

Openings brighter than ≈33 L* darken and darker ones brighten. Real frames darken more than generated frames only because they are brighter (real vs generated at equal brightness: $p = 0.80$). Different seeds of the same opening move towards a common level; an early deviation persists only 10–30%.

The prompt sets the level. We continued the same 40 openings under four prompts that differ only in their lighting description (320 videos, exploratory). Each prompt pulls the video to its own level (Fig. 2b): zero-drift levels are 42.9 [41.0, 44.7] (sunny), 33.6 (overcast; the generic prompt: 33.1), 19.1 (dusk) and 16.0 [12.5, 19.9] (night). In a joint model the lighting term is overwhelming ($p \approx 5 \times 10^{-78}$, R^2 0.82 → 0.90) and the common slope is -0.78 [-0.83, -0.72]: after 31 seconds only

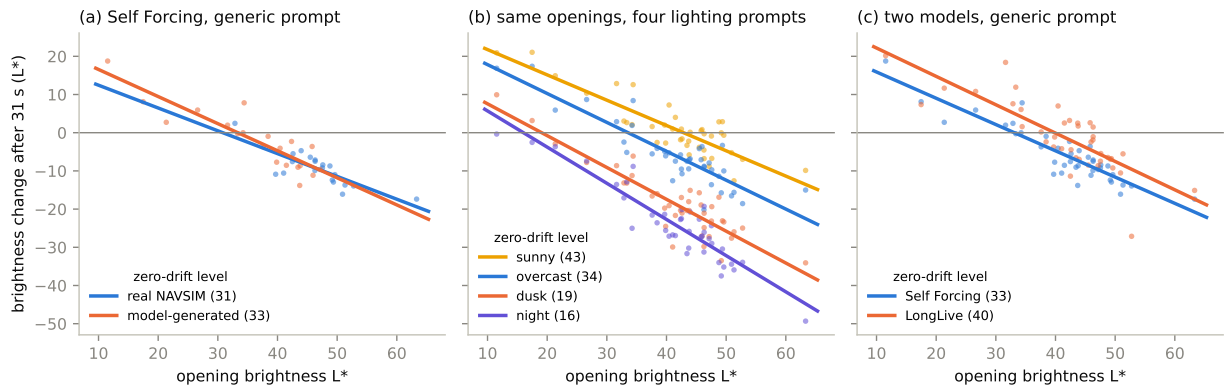


Figure 2: Brightness change after 31 s against the brightness of the held opening; one dot per opening (mean of seeds), lines are least-squares fits, legend gives the zero-drift level $-a/b$. (a) Self Forcing with the generic prompt; real and generated openings fall on the same line. (b) The same openings under four lighting prompts. (c) Self Forcing vs LongLive: same slope, different default.

about a fifth of the opening's brightness deviation is left. Darker prompts regress more completely (slope -0.95 for night vs -0.66 for sunny; interaction $p = 0.011$). For dusk and night the prompt may legitimately ask for a darker scene, so these levels describe the prompt's target, not a defect.

Second model. On LongLive the same openings regress with the same slope (-0.74 [-1.01, -0.58]; model×slope $p = 0.59$) to a brighter default (39.9 [37.5, 42.1]; $R^2 = 0.65$; Fig. 2c). LongLive seemed hardly to darken real driving frames (mean 47 L*) because they start closer to its default, not because it regresses less. Unlike Self Forcing, LongLive darkens real openings 4.3 L* more than generated ones of equal brightness ($p = 0.03$).

Interpretation. Over a long rollout the model forgets the opening and returns to what the text describes. "Colour drift" in this sense is a conflict between the visual context and the text condition that the text wins. This matters for world models that start from real camera frames and a caption.

5 A training-free correction

What failed. A statistical closed-loop brightness controller (integral control with a jump guard and a CUSUM trigger) could not tell legitimate dimming from drift: it lifted tunnels to over-exposure, and its jump guard ratcheted on its own corrections. A regression-derived "brightness direction" in latent space realised only 18% of an intended change; a direction estimated from designed per-channel perturbations realised 96%. Observational estimates gave the wrong causal answer here.

Content-aware correction. At denoising step 1 of 4, the same noisy chunk is run once more with a temporary KV cache that holds only the clean anchor frames. The difference between the per-channel spatial means (the "DC" tone of each of the 16 latent channels, per frame) of that prediction and of the real prediction is added, scaled by β. Layout and detail are untouched, and so are the main KV cache and the random stream. It costs one extra forward pass per chunk. The denoising step is a knob between drift removal and content fidelity: correcting at step 0 removes

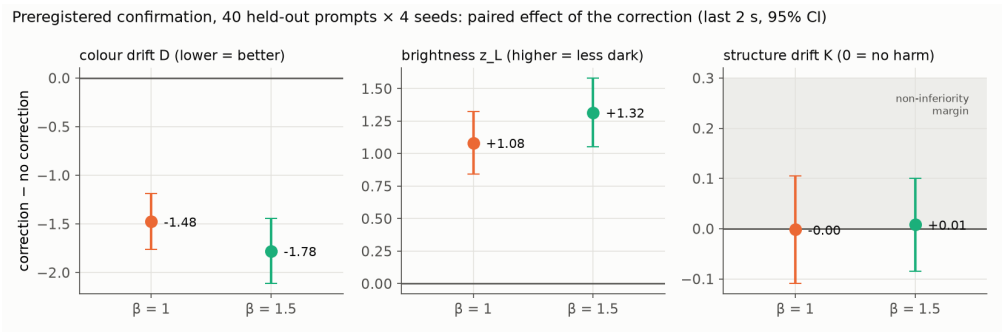


Figure 3: Preregistered confirmation of the correction on Self Forcing: paired effects on 40 held-out prompts × 4 seeds with prompt-cluster bootstrap 95% CIs.

most drift but erases legitimate transitions such as entering a tunnel (56% kept), step 1 keeps 92%, and steps 2–3 have no effect.

Confirmation (preregistered). On 40 held-out prompts × 4 seeds, on one GPU architecture, every criterion was met: D −1.48 [-1.76, -1.19] (β=1) and −1.78 [-2.11, -1.45] (β=1.5); z_L +1.08 and +1.32; K non-inferior (difference ≈ 0, CI ±0.11); held-out control prompts with legitimate brightness changes kept a median 83% / 93% of their transition. Flicker is the cost: jitter ×1.13 / ×1.67 and chunk-boundary jumps ×1.32 / ×2.40. On held real openings, the colour shift from the opening fell from 10.7 to 8.1 / 7.8 L* without more flicker.

Transfer to LongLive (exploratory). Ported without changes, the correction reduces D by 0.70 [-1.13, -0.27] on the 40 main prompts (2 seeds), keeps structure (K +0.05 [-0.10, 0.20]) and adds little flicker (×1.10). On held openings $|\Delta L_{\text{end}}|$ falls from 7.4 to 5.2 (30/40 scenes improved). The effect is about half that on Self Forcing because LongLive darkens less on these prompts (z_L −0.67 vs −2.19); β=1.5 brings no further gain.

6 Matching the prompt to the opening (preregistered)

Design. If drift is the text overriding the opening, a prompt that describes the opening should drift less. We froze the analysis, the openings and all prompts before generating any video (public commit 43bbd4c). The 40 new openings are 20 NAVSIM frames from logs not used before, chosen evenly over brightness (15.6–60.7 L*), and 20 new model-generated first frames. For each opening, Qwen2.5-VL-3B-Instruct (greedy decoding) wrote a scene phrase and a lighting sentence. Three prompts were built: the generic prompt (G); scene + lighting from the caption (M); and the generic scene with the caption's lighting sentence only (ML). G and M were run with and without the correction, ML without, on both models with 2 seeds: 800 videos, all arms of an opening on one GPU. Primary hypotheses: $|\Delta L_{\text{end}}|$ is lower for M than G (H1) and lower for M+correction than M (H2), on each model, Holm-corrected over the four tests.

Primary result: not supported. H1: −1.21 [-3.09, 0.36] on Self Forcing (Holm $p = 0.58$) and −0.19 [-2.00, 1.63] on LongLive. H2: −1.00 [-1.89, -0.12] on Self Forcing (nominal $p = 0.035$, Holm $p = 0.14$) and −0.36 [-1.48, 0.75] on LongLive. Captioning the opening with an off-the-shelf VLM does not reliably reduce brightness drift (Fig. 4a).

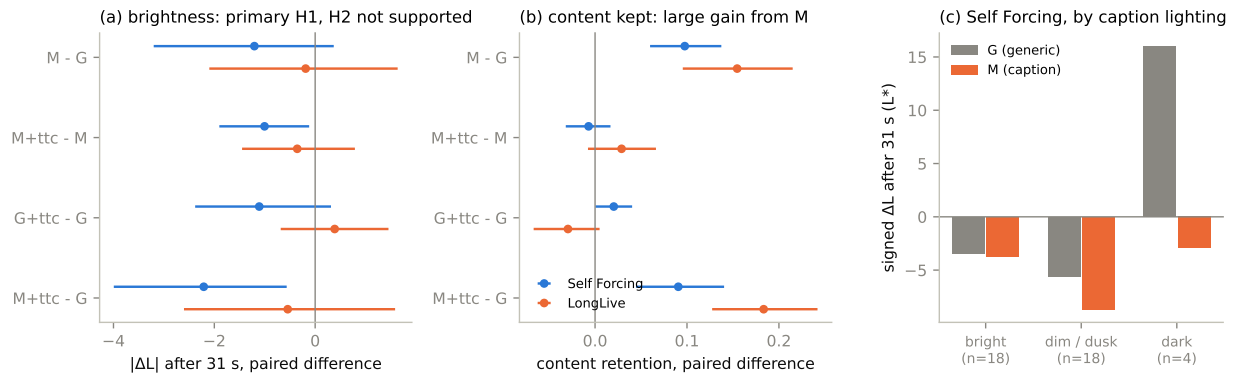


Figure 4: Matched-prompt experiment (40 new openings × 2 seeds per model). (a, b) Paired differences between arms with opening-cluster bootstrap 95% CIs. (c) Exploratory: signed brightness change on Self Forcing by the lighting words of the VLM caption.

Secondary results. The caption prompt strongly improves *content* retention, the DINOv2 cosine similarity between the last 2 seconds and the opening: +0.10 [0.06, 0.14] on Self Forcing (0.40 → 0.50) and +0.15 [0.10, 0.21] on LongLive (0.40 → 0.56) (Fig. 4b). It also lowers jitter (×0.79, ×0.71). On Self Forcing, caption prompt plus correction vs the generic prompt reduces $|\Delta L|$ by 2.21 [-3.99, -0.56] and the colour shift ΔE by 2.93 [-4.83, -1.20]. The lighting sentence alone (ML) does nothing for brightness.

Why (exploratory). Grouping openings by the lighting words of their caption shows the mechanism of §4 at work, with a miscalibrated input (Fig. 4c). The captioner called half of the normally lit frames "dimly lit, likely early morning or late evening" (mean 40 L*, up to 61). For these frames the caption prompt made Self Forcing darken *more* (−5.6 → −8.7; ML −11.2). For the 4 frames captioned as night or very dark (mean 21 L*), the generic prompt brightened them by 16 L* and the caption prompt kept them (−2.9). The prompt does set the level, but a free-form caption does not say how bright the scene is. A lighting description calibrated to measured brightness is the natural next test; with only 4 dark openings, this breakdown is a hypothesis, not a finding.

7 Calibrating the lighting description (preregistered)

Design. If the failure in §6 comes from the caption's lighting words, choosing those words from measured brightness should work. We froze the rule before generation (public commit d5e4c24). Each opening gets the Part 4 lighting prompt whose Self Forcing zero-drift level is nearest to its brightness (night 16, dusk 19, overcast 34, sunny 43; levels estimated on the earlier, independent openings). This gave 23 sunny, 10 overcast, 5 dusk and 2 night. New arms: CL (generic scene + calibrated lighting, i.e. the Part 4 prompt itself), C (the VLM scene phrase + calibrated lighting) and C+correction. They were compared with the existing G and M videos of §6. Each opening ran on the GPU architecture of its earlier arms. A rerun of G (seed 0) reproduced all 80 stored videos bit for bit, so the new and old arms are strictly paired. While preparing this run we found that 4 LongLive openings of §6 had arms split across architectures (a preempted job resumed on another GPU type). We regenerated them; the preregistered conclusions of §6 did not change. Primary hypotheses, per model and Holm-corrected: $|\Delta L_{\text{end}}|$ lower for C than M (H1) and lower for C than G (H2).

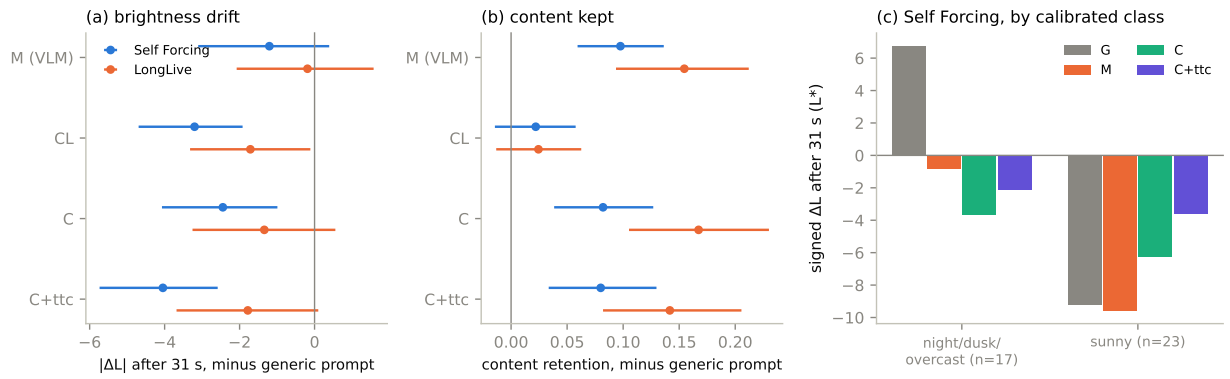


Figure 5: Calibrated lighting descriptions. (a, b) Paired differences from the generic prompt (40 openings, seed means, opening-cluster bootstrap 95% CIs). (c) Self Forcing, signed brightness change by calibrated class.

Results. H2 is supported on Self Forcing: -2.45 [-4.08, -1.04] ($9.29 \rightarrow 6.84$, Holm $p = 0.013$). H1 points the same way on both models (-1.24 and -1.15) but is not significant, and neither is H2 on LongLive (-1.34 [-3.19, 0.55]). Secondary results (Fig. 5):

- On Self Forcing, the plain calibrated prompt CL reduces $|\Delta L|$ by 3.20 [-4.69, -1.89].
- Calibrated prompt + correction vs generic reduces $|\Delta L|$ by 4.05 [-5.64, -2.54] (44%) and ΔE by 4.54 (39%). Content retention improves (+0.08) and jitter falls.
- Content retention comes from the VLM scene phrase (C vs CL: +0.06 on Self Forcing, +0.14 on LongLive), not from the lighting words.
- The Part 4 law, applied out of sample, predicts the CL outcomes on Self Forcing moderately (correlation 0.64, R^2 0.34).
- On LongLive the same predictions are biased by +4.3 L*: its defaults are higher, so dark openings were still brightened under the Self-Forcing-calibrated prompts. A calibration has to be estimated per model.

8 What did not hold

We report claims that we made at some point during the study and later revised or withdrew after further checks:

- *"MUSIQ rewards collapsed videos."* Seen in the pilot, not replicated in the main experiment: MUSIQ is simply insensitive.
- *"The sink works by stabilising the default level."* Withdrawn: the trend difference was +0.5 [-2.4, 3.5].
- *"The model follows its recent context about 75%."* True only for an artificial tone shift. For natural differences (swapping in another seed's window by KV-cache replay, bit-exact) it follows 90–100%. The sink frames have a direct effect of only 0.036, yet they reduce drift; how is not explained.

- *A dynamics model that forecasts the final brightness of a video.* Out-of-sample forecasts were weak.

9 Limitations

Both models share the Wan2.1-1.3B base. The preregistered positive results are on Self Forcing; of the two preregistered tests on both models, the first (§6) was negative and the second (§7) positive on Self Forcing only. The lighting calibration has four discrete levels and was estimated on Self Forcing; openings brighter than the sunny level (43 L^*) have no matching description. The lighting and LongLive experiments use 2 seeds. Videos are 31 seconds. Real openings are single held frames, not real video. The correction acts on colour and brightness only; structural degradation (for example sky textures late in a video) remains.

10 Related work

Self Forcing [1] and its long-video successors (Self-Forcing++, LongLive [2], Rolling Forcing, Deep Forcing [3]) use rolling windows, sinks and re-aligned positions; we measure these choices with uncertainty. Pathwise test-time correction [4] and FreqForcing [5] correct drift at inference; our correction is a simplification (DC-only transfer at one early step) studied with a preregistered confirmation. Meta-ARVDM [6] analyses error accumulation theoretically.

References

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